Hair-to-blood ratio and biological half-life of mercury: experimental study of methylmercury exposure through fish consumption in humans.

The hair-to-blood ratio and biological half-life of methylmercury in a one-compartment model seem to differ between past and recent studies. To reevaluate them, 27 healthy volunteers were exposed to methylmercury at the provisional tolerable weekly intake (3.4 μg/kg body weight/week) for adults through fish consumption for 14 weeks, followed by a 15-week washout period after the cessation of exposure. Blood was collected every 1 or 2 weeks, and hair was cut every 4 weeks. Total mercury (T-Hg) concentrations were analyzed in blood and hair. The T-Hg levels of blood and hair changed with time ($p < 0.001$). The mean concentrations increased from 6.7 ng/g at week 0 to 26.9 ng/g at week 14 in blood, and from 2.3 to 8.8 μg/g in hair. The mean hair-to-blood ratio after the adjustment for the time lag from blood to hair was 344 ± 54 (S.D.) for the entire period. The half-lives of T-Hg were calculated from raw data to be 94 ± 23 days for blood and 102 ± 31 days for hair, but the half-lives recalculated after subtracting the background levels from the raw data were 57 ± 18 and 64 ± 22 days, respectively. In conclusion, the hair-to-blood ratio of methylmercury, based on past studies, appears to be underestimated in light of recent studies. The crude half-life may be preferred rather than the recalculated one because of the practicability and uncertainties of the background level, though the latter half-life may approximate the conventional one.